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GRIDSHIELD AI — COMPLETE PROTOTYPE VISUAL EVIDENCE DOSSIER

Autonomous Purple Team SOC & Decentralized Customer DER Silicon Defense Platform

PROJECT NAME: GridShield AI PROBLEM CLUSTERS: Cluster 4 (Purple SOC) & Cluster 5 (DER Defense)
LEAD AI ENGINEER: Pulkit Agrawal CYBERSECURITY LEAD: Kabir Roy
LIVE PRODUCTION URL: https://e-on-project.vercel.app/ GITHUB REPOSITORY: https://github.com/Kabirroy12345/E.ON-Project

Executive Verification Overview:
This dossier contains direct high-resolution prototype captures of the live GridShield AI web application and SOC architecture across all 7 operational modules. Each page visualizes a core operational capability: Autonomous Red vs Blue AI Simulation (Cluster 4), Decentralized 1.14ms TinyML Edge Defense (Cluster 5), real-time 3D Holographic Weather Radar, Continuous Substation Blast Radius Prediction in response to the January 2026 Berlin Cable Bridge Arson incident, and Frugal Engineering & TCO Optimization (€0.80/device/year baseline).

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Module 1: Web3 Command Center & Tactical Radar Scope Platform Portal // Web3 Dark Cyberpunk UI
Interactive Web3 hero interface featuring hacker decode typography ([GRIDSHIELD]), 3D mouse-tracking parallax radar HUD, live
DEFCON status, and key performance badges (<140ms GNN, 800+ FPS Cortex-M4).

Route: / (Hero View)

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Module 2: 3D Holographic Substation Radar & Weather Telemetry Real-Time Cyber-Physical Weather Radar
Interactive 3D HTML5 Canvas holographic globe rendering real-time satellite orbital sweeps, ambient temperature, wind velocity, and
solar irradiance across German, Dutch, and Swedish substation nodes.

Route: / (Substation Telemetry)

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Module 3: Autonomous Defense Checkpoints & F1 Speedway Interactive Threat Mitigation Simulation
60 FPS cursor-tracking cyber speedway engine visualizing the 4 core defense checkpoints: Offensive Red Team AI, GraphSAGE GNN
Blast Radius, Zero-Telemetry Edge TinyML, and Cluster 5 Adoption.

Route: / (Cyber Speedway Circuit)

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Module 4: Security SOC Dashboard — Live Telemetry & Grid Topology Cluster 4: Improvement of Protection Systems
Live SOC operations view monitoring 9.6M virtual customer DER nodes with automated DEFCON threat gauges, real-time power network
topology graph, and automated SOAR enforcement status.

Route: /dashboard (Primary View)

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Module 5: Security SOC Analytics — Attack Curves & Incident Stream Cluster 4: Closed-Loop AI Telemetry
Real-time attack frequency vs response curve, MITRE ATT&CK; for ICS threat vector distribution, live SOC incident stream with
automated containment badges, and CVSS category breakdown.

Route: /dashboard (Incident & MITRE View)

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Module 6: Purple Team AI — 6-Stage Autonomous Hardening Loop Cluster 4: Unified Red vs Blue AI Loop
Unified 6-stage self-evolving loop (Attack -> Detect -> Analyze -> Patch -> Verify -> Learn) synchronizing continuous neural weight
updates across 14,892+ executed simulation passes.

Route: /purple-team (Loop Architecture)

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Module 7: Purple Team AI — Red Exploit vs Blue Response Engine Cluster 4: Automated Incident Hardening
Interactive attack emulation matrix featuring the January 2026 Berlin Cable Bridge Sabotage scenario, demonstrating 1.9s DPI detection
and automated firewall rule deployment with 91.6% AI confidence.

Route: /purple-team (Attack Matrix)

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Module 8: Purple Team AI — Real-Time Terminal Log & CVE Registry Cluster 4: Traceable SOC Hardening Logs
Live terminal audit stream detailing step-by-step exploit execution, lateral movement tracing, and live vulnerability registry tracking CVSS
severity scores and mitigation statuses.

Route: /purple-team (Terminal & CVE)

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Module 9: Customer DER Asset Shield — Inventory & Risk Matrix Cluster 5: Protection Schemes for

Customer-Based Assets

Decentralized protection matrix across 9.6M customer assets (EV Chargers, Heat Pumps, Solar Inverters, Home Batteries, HEMS) with
real-time risk scores and protocol vulnerability audits.

Route: /asset-shield (Asset Registry)

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Module 10: Customer DER Asset Shield — Real-Time Telemetry & Anomaly Cluster 5: 1.14ms TinyML Anomaly Detection
Live 24-hour EV charger power behavior curve and fleet anomaly scoring gauge, validating on-device zero-trust execution under
coordinated demand surge simulations.

Route: /asset-shield (Waveform Telemetry)

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Module 11: Customer DER Asset Shield — 4-Pillar Incentive Framework Cluster 5: Customer Adoption & Incentives DSO integration response architecture and 4-pillar incentive model: 4–8% dynamic grid tariff discount, up to 25% cyber insurance
discount, 100% GDPR zero-leak guarantee, and hardware warranty.

Route: /asset-shield (Incentive Model)

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Module 12: Global Market Adaptability — Cross-Border Matrix Global Scalability // US, Australia, ASEAN, EU
Cross-border grid adaptability matrix analyzing $4.63B global TAM across US NERC-CIP, Australia AEMO NEM, ASEAN Microgrids, and
EU NIS2 Article 21 with live region filters.

Route: /global-defense (Market Grid)

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Module 13: Silicon Benchmarks & IEEE 39-Bus Accuracy Objectives Hardware Specifications & Model Performance
Silicon benchmark specifications on ARM Cortex-M4 (1.14ms, 118KB RAM, 1.8mW) and Cortex-M7 with interactive benchmark runner,
paired with IEEE 39-bus accuracy targets (98.4% F1-Score, 99.1% Precision).

Route: /global-defense (Silicon Specs)

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Module 14: Architectural Comparison & Technical Verification Protocols Judge Defensibility & Technical Verification
Qualitative comparison matrix against legacy SIEMs and cloud taps, alongside expandable mathematical proofs detailing GraphSAGE
2-hop sampling (K=2, S1=25, S2=10) in <140ms.

Route: /global-defense (Defensibility Proofs)

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Module 15: Solution Architecture — Defense in Depth (Layers 1 & 2) Enterprise OT/IT Zero-Trust Architecture
Layer 1 Edge Silicon (<1.14ms, <800KB footprint) and Layer 2 Gateway Layer (10K msgs/sec, Modbus/OCPP/IEC 61850 protocol
normalizer) with 72-hour store-and-forward caching.

Route: /architecture (Layers 1 & 2)

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Module 16: Solution Architecture — Cloud GNN & SOC SOAR (Layers 3 & 4) Enterprise OT/IT Zero-Trust Architecture
Layer 3 Cloud Analytics (PyTorch GraphSAGE 2-hop GNN, STIX/TAXII threat intel fusion) and Layer 4 SOC Operations (<1.83s MTTD,
92% automated SOAR pass-thru).

Route: /architecture (Layers 3 & 4)

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Module 17: Production Tech Stack & 5-Step Data Pipeline Architecture Production Engineering // 8 Core Engines
Production stack (TinyML, Kafka 4.2 GB/s, Kubernetes k8s-v1.29, PyTorch GNN, Neo4j, TimescaleDB, OpenCTI, React D3.js) and
interactive 5-step data pipeline simulator.

Route: /architecture (Tech Stack & Pipeline)

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Module 18: Prototype vs Enterprise Stack & EU Regulatory Matrix Compliance // NIS2, GDPR, IEC 62351, IEC

62443

Direct comparison between hackathon prototype capabilities and enterprise production roadmap, paired with full coverage matrix for EU
NIS2, GDPR, IEC 62351, EN 50549, and ISO 27001.

Route: /architecture (Compliance Matrix)

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Module 19: 18-Month Scaling Roadmap & Architect's Technical Memo Program Rollout & Executive Architecture

Memo

3-phase deployment roadmap (Phase 1 Foundation Pilot, Phase 2 Scaled Rollout, Phase 3 Enterprise Maturity) with Lead AI Engineer
technical rationale on edge-first architecture.

Route: /architecture (Roadmap & Memo)

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Module 20: Impact & TCO — Market Opportunity & Interactive TCO Calculator Frugal Engineering // €0.80/device/year

Baseline

Executive market overview (€1.1B TAM, €350M SAM, 9.6M DER fleet) paired with dynamic TCO slider modeling €37.8M annual savings
and 4567% ROI at production scale.

Route: /impact (TCO Calculator)

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Module 21: Impact & TCO — GNN Defensibility & 5-Year ROI Projection Financial Validation & Sub-2s GNN Scaling
Detailed mathematical proof of sub-2s GNN scaling across 9.6M nodes, 5-year cumulative ROI projection curve, and itemized cost
breakdown donut visualization.

Route: /impact (ROI Projections)

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Module 22: Impact & TCO — EU Country SAM & Indian Innovation Strengths Criteria for All Challenges // Indian Innovation
Country-by-country European addressable market distribution alongside Bangalore R&D; Labs spotlight on constraint-driven engineering
and legacy hardware retrofits.

Route: /impact (Global Expansion & India
Strengths)

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Module 23: Impact & TCO — Frugal Mindset & Quantified Impact Metrics Quantified Impact // 450M+ EU Households
Frugal innovation pillars, Lead Architect verified benchmark quote with official badge, and quantified impact metrics: 450M+ protected
households, 87% faster triage, and €12.6M annual savings.

Route: /impact (Frugal Engineering & Impact)

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